MOHAMMED ARMAAN

UAV Design Engineer | Fixed-Wing Aircraft | CFD & FEA | Composite and 3D-Printed Airframes
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PROFESSIONAL SUMMARY

Aerospace Engineering with hands-on UAV design experience across aerodynamic sizing, stability analysis, CFD and FEA, composite and 3D-printed fabrication, and ArduPilot flight-controller integration. Led the design of a variable-stability fixed-wing UAV and presented the work as first author at ASET 2026. Actively working on new skills and research and seeking a UAV design role to make a real impact on the industry.

EDUCATION

B.Tech. Aerospace Engineering, Amity University Dubai, UAE | CGPA: 9.68 | Sep 2022 – Jun 2026

ENGINEERING PROJECTS

Variable-Stability Fixed-Wing UAV with Telescoping Tail Boom (Team Lead) | 1.8 m span, 3.6 kg AUW

- Led a multi-disciplinary team to design a UAV whose actuated tail boom varies horizontal tail volume coefficient (0.36 to 0.44) in flight; an ESP32-controlled battery carriage offsets the shift, limiting CG movement to 6.16 mm.
- Ran 3D CFD in ANSYS Fluent (k-omega SST, 3.6M cells, 18-point sweep) and corrected a destabilising pitching-moment slope by re-lofting the fuselage; pitch stiffness dCm/dα steepened from -0.0031 to -0.0049 per degree as the tail extends; L/D 8.51 extended vs 7.61 retracted (15 m/s, 0°).
- Sized wing and tail using XFLR5 and Excel; ran SolidWorks FEA at 3g limit load: minimum factor of safety 3.69 (wood wing spars) and 21.7 (carbon fibre tail boom).
- Built a flight-ready foam-core, fibreglass, carbon and wood airframe (ArduPlane, 25 Hz data logger); ground taxi-tested to 5 m/s; flight trials pending GCAA airspace clearance. Copyright application filed.

RC Cessna 172, 3D-Printed Flying Model | 1:8 scale, 1.42 m span, 3.6 kg AUW, 4S LiPo

- Designed and fabricated a flying model with spiral-vase PETG wings and a three-piece fuselage; flight-tested, with short endurance traced to high structural weight.

Hybrid-Power Cloud-Seeding UAV (Conceptual Design) | 2.4 m span fixed-wing

- Led preliminary sizing and concept development for UAE weather conditions; proposed a hybrid propulsion architecture (IC engine + dual electric) and began prototype fabrication (carbon fibre and aluminium fuselage, 3D-printed wing panels).

Lunar Rover Prototype (Conceptual) | 6-DoF arm, 6WD, custom frame

- Designed a robotic arm integrated into an articulated rover; programmed an ESP32-based multi-mode control system; modelled and fabricated PETG and carbon-fibre structures.

PROFESSIONAL EXPERIENCE

DfAM and FDM Operations Intern, Ultratec 3D Printing, Dubai, UAE | Jul 2026 – Aug 2026

- Repaired STL meshes and partitioned large parts with custom mechanical joints (Lu Ban, Voxeldance); reconstructed 3D scans (EinScan Rigil, IR and laser) into parametric CAD models.
- Assisted in drafting GD&T-compliant production drawings; managed multi-printer FDM operations (Bambu Lab X1C, H2D), tuning print parameters to reduce failures and downtime.

Aircraft Maintenance Intern, Egnatia Aviation Ltd, Kavala, Greece | Jul 2024 – Aug 2024

- Assisted with scheduled maintenance (gear, engines), aircraft balancing, ground-run diagnostics, carburettor synchronisation.

PUBLICATIONS

- Armaan, M., Patwa, H. K., Prakasan, A. P., Raj, S., Pillai, S. **Development of a Telescoping Boom Variable Stability Unmanned Aerial Vehicle.** Submitted to and presented at ASET 2026. Pending publishing.
- Armaan, M., Ntantis, E. (2025). **Performance Simulation of a Micro Gas Turbine Using Various Sustainable Aviation Fuels.** Preprint, Research Square. DOI: 10.21203/rs.3.rs-7528156/v1.

AWARDS

2nd Prize, PlanetX Lander Challenge, UAE Space Agency (Dec 2025): prototyped a shock-absorbing 1U CubeSat lander that survived a 2 m drop test.

1st Prize, Third Emirates Robotic Competition, AUS (Apr 2025): designed and integrated an 800 mm RC catamaran hull for autonomous water cleaning.

TECHNICAL SKILLS

UAV Design and Analysis: Aerodynamic sizing, stability and control, CFD (ANSYS Fluent), FEA (SolidWorks Simulation), XFLR5, Excel
CAD and Drafting: SOLIDWORKS, NX, CATIA, AutoCAD, ASME Y14.5 GD&T, DXF/DWG, eDrawings, 3D PDF
Fabrication: FDM and resin 3D printing, foam-core composites, hand layup, hot-wire cutting
Flight Control and Avionics: ArduPilot/ArduPlane, PX4, INAV, Mission Planner, CubeOrange, F405, SBUS, telemetry
Embedded and Electronics: ESP32, Raspberry Pi, Python, C++, PID control loops, power electronics, sensor integration and filtering

CERTIFICATIONS

- SOLIDWORKS Certified Professional (CSWP) – Design; SOLIDWORKS Flow Professional
- CSWPA (Advanced): Surfacing, Mechanical Design Validation (MBD), Weldments, Sheet Metal, Drawings
- Hobby-Class Fixed-Wing UAV Pilot Certification, Al Lisaili RC Flying Club, Dubai